

Section A: Basic Concepts

1. What is a socket?

- A) A physical device on a network card
- B) A communication endpoint identified by a socket descriptor
- C) A hardware protocol
- D) A routing algorithm

Answer: B

2. A socket address is a combination of:

- A) MAC address + port
- B) IP address + port number
- C) IP address + subnet mask
- D) MAC address + IP address

Answer: B

3. IPv4 address length is:

- A) 16 bits
- B) 32 bits
- C) 64 bits
- D) 128 bits

Answer: B

4. Which notation is used for IPv4 addresses?

- A) Binary notation
- B) Dotted decimal notation
- C) Hexadecimal notation
- D) Octal notation

Answer: B

5. Example of IPv4 address:

- A) 2001:db8::1
- B) 138.23.44.2
- C) 10101010
- D) FF:FF:FF:FF

Answer: B

Section B: Socket Types

6. Datagram sockets use:

- A) TCP
- B) UDP
- C) HTTP
- D) FTP

Answer: B

7. Stream sockets use:

- A) TCP
- B) UDP
- C) ICMP
- D) DNS

Answer: A

8. UDP provides:

- A) Reliable communication
- B) Connection-oriented service
- C) Connectionless communication
- D) Guaranteed delivery

Answer: C

9. TCP provides:

- A) Unreliable communication
- B) Reliable communication
- C) Broadcast communication
- D) Datagram communication

Answer: B

Section C: Port Numbers

10. Total number of ports available:

- A) 1024
- B) 4096
- C) 65536
- D) 10000

Answer: C

11. Well-known ports range:

- A) 0 – 1023
- B) 1024 – 49151
- C) 49152 – 65535
- D) 1 – 500

Answer: A

12. Dynamic ports range:

- A) 0 – 1023
- B) 1024 – 49151

- C) 49152 – 65535
- D) 200 – 400

Answer: C

13. HTTP uses port:

- A) 21
- B) 25
- C) 80
- D) 443

Answer: C

Section D: Byte Ordering

14. Internet uses which byte order?

- A) Little endian
- B) Big endian
- C) Random order
- D) Host order

Answer: B

15. Network byte order is:

- A) Little endian
- B) Big endian
- C) Reverse endian
- D) Binary endian

Answer: B

16. Function used to convert host to network for 32-bit values:

- A) htons()
- B) ntohs()
- C) htonl()
- D) ntohl()

Answer: C

17. Function used to convert network to host for port numbers:

- A) ntohs()
- B) htons()
- C) htonl()
- D) ntohl()

Answer: A

Section E: Socket Address Structure

18. IPv4 socket address structure name:

- A) sockaddr
- B) sockaddr_in
- C) sockaddr_tcp
- D) sockaddr_ipv4

Answer: B

19. Header file containing sockaddr_in:

- A) stdio.h
- B) sys/socket.h
- C) netinet/in.h
- D) stdlib.h

Answer: C

20. Field storing port number in sockaddr_in:

- A) sin_family
- B) sin_port
- C) sin_addr
- D) sin_len

Answer: B

21. Field storing IPv4 address:

- A) sin_family
- B) sin_port
- C) sin_addr
- D) sin_zero

Answer: C

22. Value of sin_family for IPv4:

- A) AF_INET
- B) AF_IP
- C) AF_TCP
- D) AF_NET

Answer: A

Section F: Generic Socket Address

23. Generic socket structure name:

- A) sockaddr
- B) sockaddr_gen
- C) socket_addr
- D) sockaddr_in

Answer: A

24. Field indicating address family:

- A) sa_data
- B) sa_family
- C) sa_len
- D) sa_addr

Answer: B

Section G: Process

25. A process is:

- A) Stored program
- B) Executing instance of a program
- C) Hardware interrupt
- D) Network protocol

Answer: B

Section H: Client Server Model

26. In client-server model:

- A) Server sends request
- B) Client sends request
- C) Both send requests
- D) Router sends request

Answer: B

27. Server does what?

- A) Requests service
- B) Responds to client requests
- C) Sends broadcast
- D) Creates clients

Answer: B

Section I: UDP Characteristics

28. UDP is:

- A) Connection oriented
- B) Connectionless
- C) Reliable
- D) Encrypted

Answer: B

29. UDP guarantees delivery?

- A) Yes
- B) No

Answer: B

30. UDP guarantees packet order?

- A) Yes
- B) No

Answer: B

Section J: Socket API Syntax

31. Syntax of socket creation:

A)

```
int socket(int family, int type, int protocol)
```

B)

```
socket(int type, int protocol)
```

C)

```
create_socket()
```

D)

```
socket_create()
```

Answer: A

32. Function used to bind socket to IP and port:

A) connect()

B) bind()

C) listen()

D) accept()

Answer: B

33. Syntax of bind:

A)

```
int bind(int sockfd, struct sockaddr*, int addrlen)
```

B)

```
bind(socket, port)
```

C)

```
bind(sock, ip)
```

D)

```
bind(port)
```

Answer: A

Section K: TCP Server Functions

34. Function used to wait for incoming connections:

- A) accept()
- B) connect()
- C) listen()
- D) recv()

Answer: C

35. Syntax of listen:

```
int listen(int sockfd, int backlog)
```

What does backlog represent?

- A) Number of sockets
- B) Size of waiting connection queue
- C) Number of ports
- D) Packet size

Answer: B

36. Function used to accept client connection:

- A) connect()
- B) accept()
- C) recv()
- D) send()

Answer: B

37. `accept()` returns:

- A) IP address
- B) Port number
- C) New socket descriptor
- D) Packet size

Answer: C

Section L: Client Function

38. Client connects to server using:

- A) `listen()`
- B) `bind()`
- C) `connect()`
- D) `accept()`

Answer: C

39. Syntax of `connect`:

```
int connect(int sockfd, struct sockaddr* servaddr, int addrlen)
```

Answer: Correct syntax.

Section M: TCP Properties

40. TCP connection establishment uses:

- A) 2-way handshake
- B) 3-way handshake
- C) 4-way handshake
- D) 5-way handshake

Answer: B

41. TCP connection termination uses:

- A) 2-way handshake
- B) 3-way handshake
- C) 4-way handshake
- D) 1-way handshake

Answer: C

Section N: True/False

42. UDP guarantees reliable delivery.

Answer: False

43. TCP is connection-oriented.

Answer: True

44. Socket descriptor is a non-negative integer.

Answer: True

45. Servers usually run continuously in background.

Answer: True

46. Client must know server address before connection.

Answer: **True**

Section O: Syntax MCQs

47. Which creates a TCP socket?

A)

```
socket(AF_INET, SOCK_STREAM, 0)
```

B)

```
socket(AF_INET, SOCK_DGRAM, 0)
```

C)

```
socket(AF_INET, RAW, 0)
```

D)

```
socket(AF_TCP)
```

Answer: **A**

48. Which creates a UDP socket?

A)

```
socket(AF_INET, SOCK_STREAM, 0)
```

B)

```
socket(AF_INET, SOCK_DGRAM, 0)
```

C)

```
socket(AF_INET6, SOCK_STREAM, 0)
```

D)

`socket(AF_TCP)`

Answer: B

49. Correct IPv4 family constant:

- A) AF_TCP
- B) AF_IPV4
- C) AF_INET
- D) AF_NET

Answer: C

50. Socket descriptor returned on error:

- A) 0
- B) -1
- C) 1
- D) NULL

Answer: B